

Question 3. Scalable Decision Trees, Random Forests, and Gradient Boosting [10 marks, set by Robert Loftin]

Prerequisites: Before attempting this question, you are expected to have completed Lab 5.

Data:

In this question, you will use supervised classification algorithms to identify [Higgs bosons](#) from particle collisions, like those produced by the [Large Hadron Collider](#). In particular, you will use the [HIGGS dataset](#).

Use **wget** to download the data using the direct link:

<http://archive.ics.uci.edu/ml/machine-learning-databases/00280/HIGGS.csv.gz>

You will then need to unzip the dataset first. For this purpose, you can use a tool like **gzip** or **unzip**.

Task:

You must use a single Slurm batch script to run your code for Q2 on Stanage for this question. Your batch script should request 10 CPU cores in total, and 8 GB of memory per core.

In part A of this question, you will apply Random Forests and Gradient Boosted Trees to a subset of the dataset, while in part B you will apply the same methods to the full dataset. To evaluate the performance of each model, you will use the binary classification accuracy.

A.

Use Spark [pipelines](#), [parameter grids](#), and [cross-validation](#) to find the best configuration of parameters for both the Random Forest and Gradient-Boosted Tree classification models on the HIGGS dataset.

1. To find the best configuration of the parameters for each of the models, you must first sample 2% of the data instances. Using only this small subset will keep your computation time manageable. *Hint: if you use [Stratified Sampling](#) to ensure a balance*

between positive and negative instances when sampling this random subset of data, you will likely find better parameter values.

2. You must optimize three configuration parameters each for the Random Forest and GBTs. You may choose any three parameters to optimize for either of the models. You must optimize over a sensible **grid of possible combinations** of values for these three parameters, with at least three possible values for each of these parameters. Use k=3-fold cross-validation to evaluate the performance of each parameter set. Make sure to include the best parameters found for each algorithm in your report. (6 marks)
3. Split the smaller dataset you used for parameter tuning (the 2% of the full dataset you sampled above) into training and test sets. Use the same training-test split to compare the performance of Random Forests (with its best choice of hyperparameters) with that of Gradient Boosted Trees (with its best choice of hyperparameters). Include the performance of each algorithm on both the training and test data sets in your report. (2 marks)

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B.

Once you have found the best parameter configurations for both algorithms using the smaller subset of the data, use the full dataset to compare the performance of Random Forests and Gradient Boosted Trees.

1. Split the full dataset into training and test sets. You will use the same train-test split to evaluate both algorithms. Specify in your report how large the training and test sets are. We will check your code to see that it produces splits of the sizes given in your report. (1 mark)
2. Use the best parameters found for each algorithm (based on the smaller dataset used in the previous step) to fit each model to the training set, and evaluate this model on the test set. In your report, compare performance on both the training and test sets between the two models. (1 mark)

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